A

```
'vedir': {
    '_default': 1.0,
    '_display_name': 'VEGF Diffusion Rate',
    '_category': 'Growth Factors',
    '_subcategory': 'VEGF Dynamics',
    '_expert_level': 'intermediate',
    '_unit': 'pixels²/MCS',
    '_dimension': 'diffusivity',
    '_mathematical_range': (0.0, 10.0),
    '_physiological_range': (0.1, 2.0),
    '_recommended_range': (0.5, 1.5),
    '_range_evidence': {
        'mathematical': {
            'basis': 'numerical_stability',
            'refs': [],
            'note': 'The explicit finite-difference step is stable only for D·Δt/Δx² ≤ 0.25 in 2D; DiffusionS […]',
        },
        'physiological': {
            'basis': 'literature',
            'refs': ['merks2006', 'merks2008'],
            'note': '0.1–2 site²/MCS brackets the Merks value (0.75 site²/MCS at 2 μm, 30 s). Free VEGF-A165  […]',
        },
        'recommended': {
            'basis': 'exploratory',
            'refs': [],
            'note': 'Presets 0.5 / 1.0 / 1.5 / 2.0. The gradient length L = √(D/ε) is what matters: with ε =  […]',
        },
    },
    '_ontology_term': 'CBO:0000090',
    '_dependencies': {
        'veder': 'Balance D/k determines gradient length scale: λ ~ sqrt(D/k)',
        'vesec': 'High diffusion smooths out discrete secretion sources',
    },
    '_expected_behavior': {
        'low_value': 'Steep local gradients, localized sprouting',
        'high_value': 'Shallow global gradients, diffuse sprouting pattern',
    },
    '_reference_paper': {
        'source': 'This model (angiogenesis_process.py)',
        'value': 1.0,
        'notes': 'Merks et al. (2006, 2008) use D = 1e-13 m²/s for the chemoattractant in the ECM, deliber […]',
    },
    '_measurable': True,
    '_experimental_assay': 'Fluorescence recovery after photobleaching (FRAP)',
    # … 17 further fields omitted (_updater, _emit, _description, _long_description, …)
}
```

B

VEGF Diffusion Rate 1.0 i ✓

vedir = 1: within every range

VEGF Diffusion Rate 3.0 i ⚠

vedir = 3: outside the supported range: warning
WARNING: Value 3.0 outside physiological range [0.1, 2.0]

VEGF Diffusion Rate 12.0 i ✗

vedir = 12: outside the admissible range: error
ERROR: Value 12.0 outside stable range [0.0, 10.0]. Simulation may fail!

C

VEGF Diffusion Rate

Ontology term: CBO:0000090

Valid Ranges

- **Physiological Range:** [0.1, 2.0] pixels²/MCS
  Basis: published evidence. Source: Merks RMH, Brodsky SV, Goligorsky MS, Newman SA, Glazier JA. Cell elongation is key to in silico replication of in vitro vasculogenesis and subsequent remodeling. Dev Biol. 2006;289(1):44–54. (doi:10.1016/j.ydbio.2005.10.003); Merks RMH, Perryn ED, Shirinifard A, Glazier JA. Contact-inhibited chemotaxis in de novo and sprouting blood-vessel growth. PLoS Comput Biol. 2008;4(9):e1000163. (doi:10.1371/journal.pcbi.1000163) — Merks et al. 2006, Results ("diffuses slowly at about 1e-13 m² s⁻¹") and Discussion (VEGF165 ≈ 1e-11 m² s⁻¹), p. 50. 0.1–2 site²/MCS brackets the Merks value (0.75 site²/MCS at 2 μm, 30 s). Free VEGF-A165 (≈1e-11 m²/s ≈ 75 site²/MCS) lies far above the band and, per Merks 2006, does not produce stable networks in this mechanism — the band therefore encodes the paper's conclusion that the morphogen must be ECM-bound and slow.
- **Recommended Range:** [0.5, 1.5] pixels²/MCS
  Basis: exploratory, not evidence-based. Locator: _presets of this parameter in AngiogenesisProcess.ports_schema(). Presets 0.5 / 1.0 / 1.5 / 2.0. The gradient length L = √(D/ε) is what matters: with ε = 0.3 the band gives L ≈ 1.3–2.2 sites, much shorter than Merks' 5 sites.
- **Mathematical Range:** [0.0, 10.0] pixels²/MCS
  Basis: numerical stability of the implementation. Locator: DiffusionSolverFE, explicit scheme. The explicit finite-difference step is stable only for D·Δt/Δx² ≤ 0.25 in 2D; DiffusionSolverFE subdivides the MCS to stay under that limit, so larger D costs run time rather than failing outright. 10 site²/MCS is the modeler's cap (40 sub-steps per MCS). D < 0 is meaningless.

Evidence Basis

- **How the value was set:** estimated
- **Directly measurable:** yes
- **Experimental assay:** Fluorescence recovery after photobleaching (FRAP)
- **Parameter type:** kinetic
- **Scope:** model parameter

Expected Behavior

- **Low values:** Steep local gradients, localized sprouting
- **High values:** Shallow global gradients, diffuse sprouting pattern

Parameter Dependencies

- **VEGF Decay Rate:** Balance D/k determines gradient length scale: λ ~ sqrt(D/k)
- **VEGF Secretion Rate:** High diffusion smooths out discrete secretion sources

D

parameter_audit.json (abridged) — run with vedir = 3

```
{
  "vedir": {
    "value": 3.0,
    "default": 1.0,
    "unit": "pixels²/MCS",
    "changed_from_default": true,
    "ranges": {
      "mathematical": [0.0, 10.0],
      "physiological": [0.1, 2.0],
      "recommended": [0.5, 1.5]
    },
    "position": {
      "mathematical": "within",
      "physiological": "above",
      "recommended": "above"
    },
    "credibility": "extrapolation",
    "range_evidence": {
      "physiological": {
        "basis": "literature",
        "refs": ["merks2006", "merks2008"]
      }
    }
  }
}
```

E

EGF Diffusion Rate (Superficial) (V-Cornea)

Units & Conversion

- **Value (physical):** 13.3333 μm²/min
- **Simulation value:** 20 voxel^2/MCS
- **Conversion rule:** display = simulation × 0.666667 (1 voxel^2/MCS = 0.666667 μm²/min)

GUI / Implementation

- Displayed in the GUI as **13.3333 μm²/min**, converted from the stored simulation value.
- Written to `Parameters.py` as `EGF_SUPERDiffCoef=20.0` (CC3D-native units — the value the model actually runs).
- The EGF field's per-region diffusion is defined in CC3DML and overwritten at runtime: `ConstraintInitializerSteppable.start()` runs `get_xml_element('EGF_Coef_SUPER').cdata = self.EGF_SUPERDiffCoef`, so the `Parameters.py` value (voxel^2/MCS) wins over the XML default.

Governing Equations

EGF Reaction-Diffusion Equation · Eq. 9-10; S5 Text

$$\frac{\partial c_{EGF}}{\partial t} = D_{EGF}(x,y)\,\nabla^2 c_{EGF} - k_{d_{EGF}}\,c_{EGF} + \Omega_{EGF}(x,y,t)$$

PDE for the EGF field: cell-type-dependent diffusion, global decay, tear source.

Other parameters in this equation: EGF Global Decay Rate · EGF Secretion Rate (Tear) · Basal EGF Uptake Rate · Stem EGF Uptake Rate · Superficial EGF Uptake Rate · Wing EGF Uptake Rate

Refs: Kuo CY (2016), Van Itallie CM (2008), Chan KY (1991)